Geometry & Trigonometry

Area and perimeter of basic shapes

1 Find the area of a triangle with base 12 and height 7.

2 Find the area and circumference of a circle with radius 5, in terms of π.

The Pythagorean theorem

3 A right triangle has legs 6 and 8. Find the length of the hypotenuse.

4 A ladder 13 feet long leans against a wall, with its base 5 feet from the wall. How high up the wall does the ladder reach?

Volume and surface area of solids

5 Find the volume of a rectangular prism with dimensions 4 by 5 by 6.

6 Find the volume of a cylinder with radius 3 and height 10, in terms of π.

Similar triangles and proportional reasoning

7 Triangle ABC is similar to triangle DEF, with $AB = 6$, $DE = 9$, and $BC = 8$. Find EF.

8 A tree casts a shadow 20 feet long, while a 6-foot person standing nearby casts a shadow 4 feet long. Find the height of the tree using similar triangles.

Right triangle trigonometry

9 In a right triangle, the angle θ has an opposite side of length 5 and hypotenuse 13. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

10 A 20-foot ladder leans against a wall, making a 60° angle with the ground. Find the height the ladder reaches on the wall, to the nearest tenth.

Angle relationships and circle theorems

11 Two parallel lines are cut by a transversal. If one angle measures 65°, find the measure of its corresponding angle and its co-interior (same-side interior) angle.

12 A central angle in a circle measures 80°. Find the measure of the inscribed angle that subtends the same arc.

The unit circle and radian measure

13 Convert 120° to radians, in terms of π.

14 Find the exact value of $\cos\left(\frac{\pi}{3}\right)$ using the unit circle.

Applying geometry and trigonometry to real-world contexts

15 A surveyor stands 100 feet from the base of a building and measures the angle of elevation to the top as 35°. Find the height of the building, to the nearest foot.

Composite area and shaded regions

16 A square with side length 10 has a circle of radius 4 cut out of its center. Find the remaining area, in terms of π.

17 Find the area of a trapezoid with parallel sides 6 and 10, and height 5.

Coordinate geometry: distance and midpoint

18 Find the distance between the points $(1, 2)$ and $(7, 10)$.

19 Find the midpoint of the segment connecting $(-3, 5)$ and $(9, -1)$.

Equation of a circle

20 Write the equation of a circle with center $(2, -3)$ and radius 5.

21 Find the center and radius of the circle $(x+1)^2 + (y-4)^2 = 36$.

Arc length and sector area

22 A circle has radius 6. Find the arc length of a 60° sector, in terms of π.

23 Find the area of a 90° sector of a circle with radius 8, in terms of π.